Does the Slope of the US Yield Curve Predict Six-Month-Ahead US Equity Returns? A Predictive Regression with Overlap-Robust Inference

SIQuant Research[1]

[1]Hypothesis × Quant

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Abstract

Hypothesis: H_0: the 10y–3m spread has no predictive power for 126-session-ahead S&P 500 returns; H_1: the slope coefficient is positive
Data: 2,387 daily observations, 2016-10-24 to 2026-04-24; 614 sessions with an inverted curve
Method: OLS with Newey–West HAC errors (126 lags) and a 2,000-draw circular-shift placebo; ACCEPTED only if both one-sided $p < 0.05$
Results: $\beta = -0.0262$ (HAC $t = -1.82$, $p = 0.9659$; placebo $p = 0.9250$), $R^2 = 0.074$.
Research decision: **REJECTED**

1 Design

The regressor is the 10-year minus 3-month Treasury spread observed at the close of day t; the outcome is the log return of the S&P 500 from the close of t to the close of $t + 126$, so there is no look-ahead [Estrella and Mishkin, 1998]. Overlapping outcomes induce strong serial correlation, therefore HAC errors with a lag equal to the horizon are used and a circular-shift placebo provides a non-parametric reference distribution for the t-statistic.

2 Results

Mean forward return is 8.33% after inverted-curve sessions (614 sessions) and 5.63% otherwise.

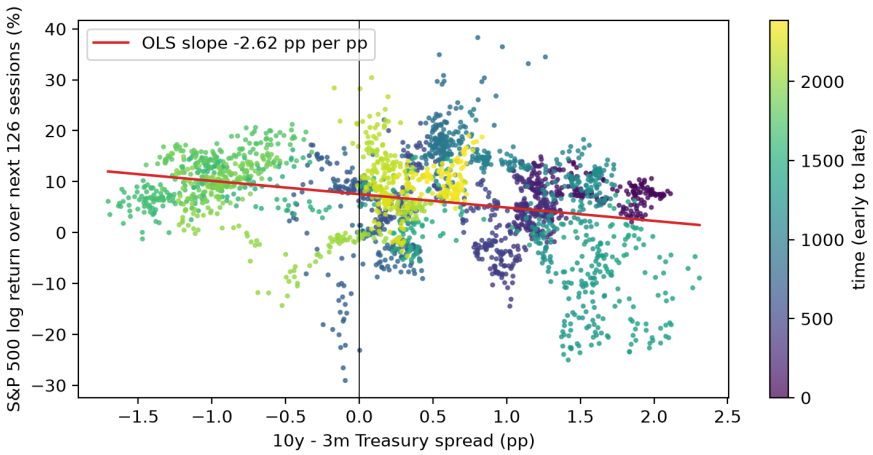


Figure 1: Six-month-ahead S&P 500 log return against the 10y–3m spread.

3 Limitations

The sample covers one full cycle with only two inversion episodes, so the effective number of independent observations is far smaller than the number of rows; the placebo accounts for this only partly. The spread is a market series and is not revised.

References

Arturo Estrella and Frederic S Mishkin. Predicting us recessions: Financial variables as leading indicators. *Review of Economics and Statistics*, 80(1):45–61, 1998.